

AI-Powered Financial Fraud Detection System

One-page project brief for internship showcase (Java Backend / Full-Stack).

Project Summary

Built a secure, production-inspired fraud detection platform using Java and Spring Boot. The system authenticates users with JWT, scores transactions using an AI-style rules engine, stores decision history in a database, and provides a professional analyst dashboard.

Key Features

- JWT authentication and role-based access control (ADMIN, ANALYST).
- Fraud scoring engine using transaction amount, velocity, account age, chargeback history, geo/IP risk signals.
- Risk outputs: score (0-100), risk level (LOW/MEDIUM/HIGH), suspicious flag, triggered rules, recommendation.
- Database-backed persistence for users and fraud assessment history.
- Admin-only history endpoint for auditability and analyst workflow review.
- Modern browser dashboard with moving visuals: live radar, dynamic risk meter, animated risk donut, and styled history table.

Tech Stack

Java 17, Spring Boot 3, Spring Security, JWT (jjwt), Spring Data JPA, H2, PostgreSQL, Maven, JUnit 5, Docker, Docker Compose, HTML/CSS/JavaScript.

API Highlights

- POST /api/v1/auth/login - issues JWT token.
- POST /api/v1/fraud/assess - secured endpoint for ADMIN and ANALYST.
- GET /api/v1/fraud/history - secured endpoint for ADMIN only.

Deployment and Demo

- Local run: `mvn spring-boot:run`
- Containerized run: `docker compose up --build`
- Default users: admin/admin123 (ADMIN), analyst/analyst123 (ANALYST)
- Dashboard URL: `http://localhost:8080`

Resume Value

Demonstrates secure API development, backend architecture, data persistence, RBAC, containerization, practical fraud-domain problem solving, and professional UI/UX presentation in a single project.